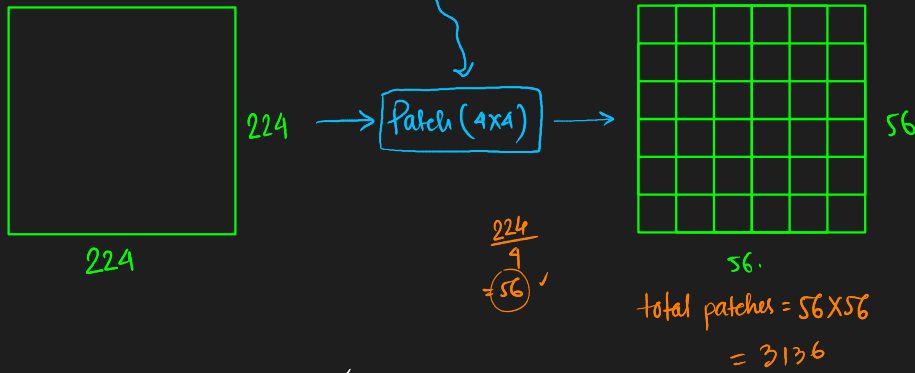
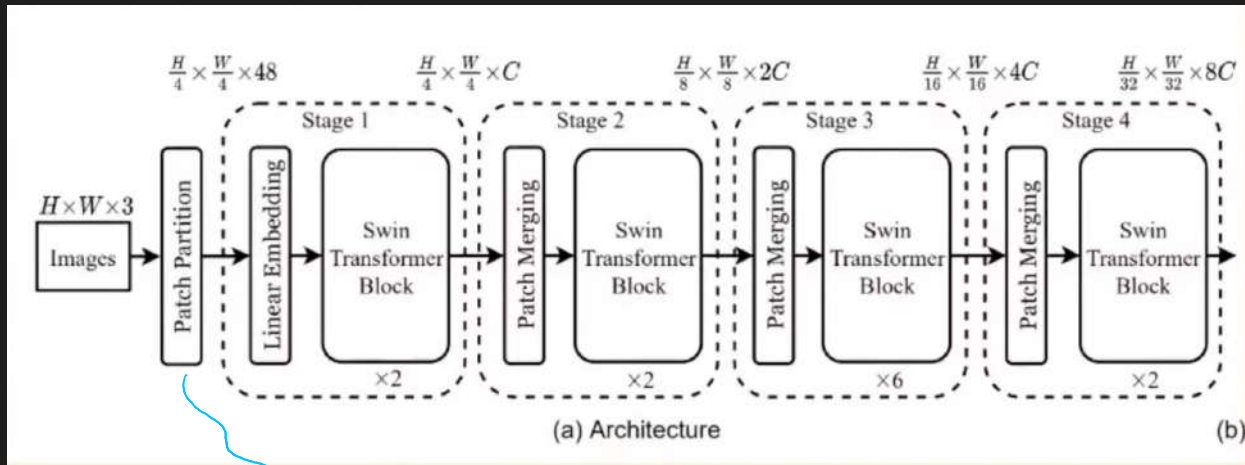
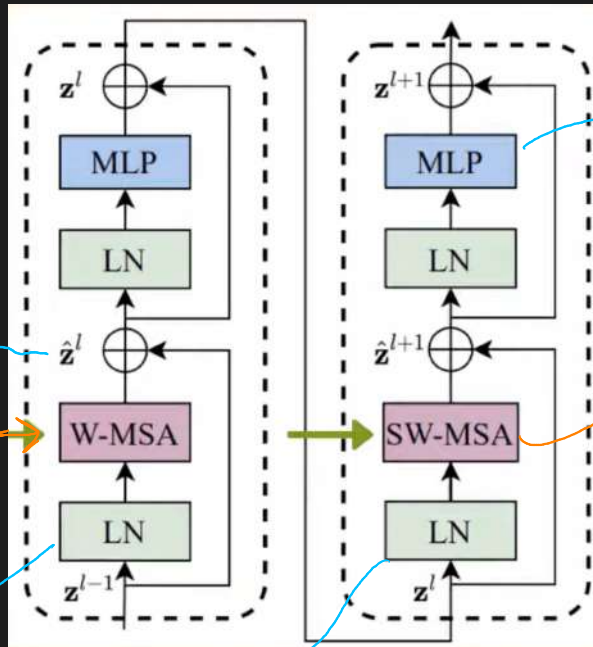


Swin Transformer



consecutive
2 Swin Transformer
Blocks



Add
Residual

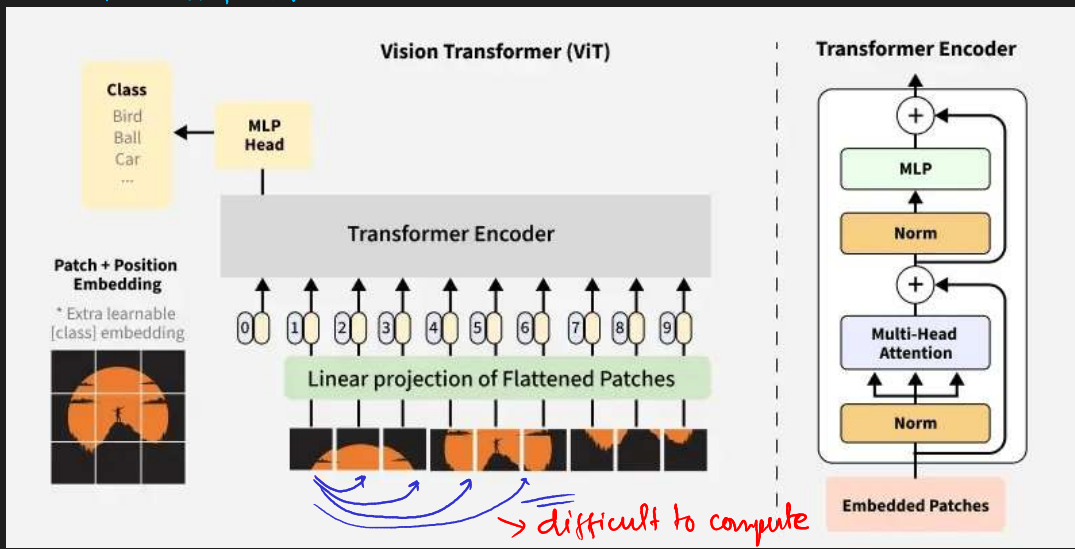
Window-
Multi Head Self
-Attention.

layer
Normalizn

Multi Layer
Perception.

Shifted Window
Multi Head SA.

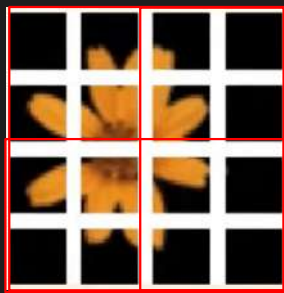
Vision Transformer →



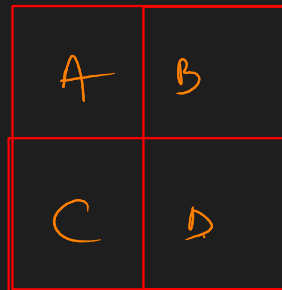
Swin Transformer

Window based MSA

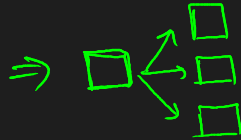
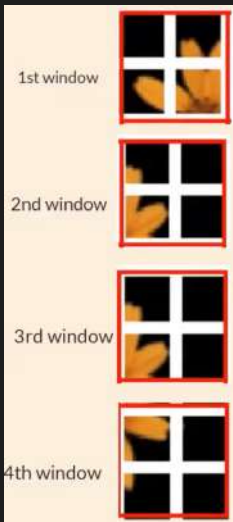
(W-MSA)



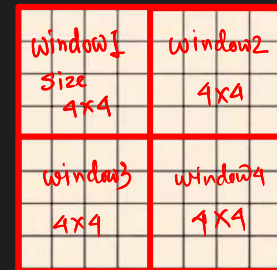
total 4 windows.



2x2 patch.



8x8 Feature Map



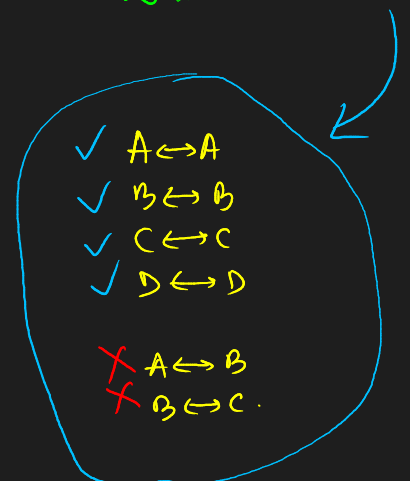
- Only provide relations within the window.
- Cannot provide relation b/w 2 windows.



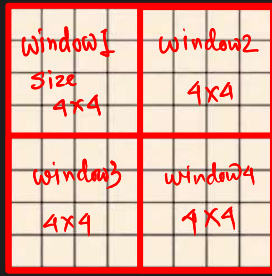
To overcome this :

SW-MSA

↓
This can provide relation b/w 2 windows.

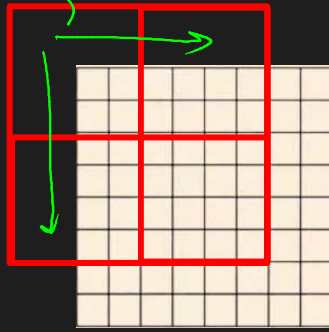


SW-MSA



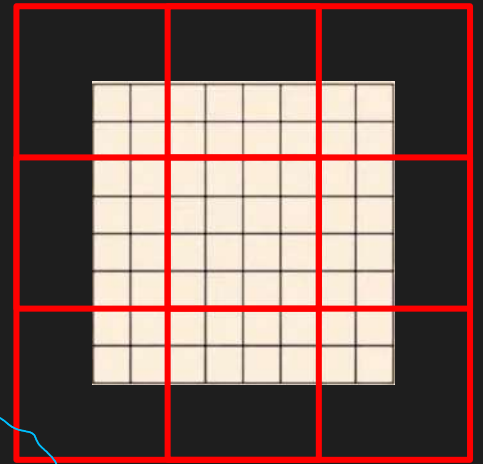
4 windows.

blank spaces.



Shift = 2.

9 windows



windows ↑↑
8 blank spaces.

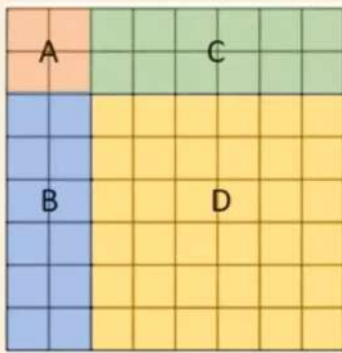
Computationally Expensive.

Cyclic Shift.

(shifted images are no longer aligned in 4x4 block - so, wrap them)

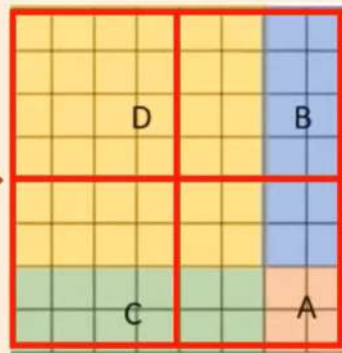
just like circular array.

Shifted Window Self Attention.

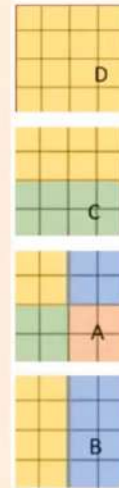


original patches

Cyclic shift



shifted patches



Masked MSA

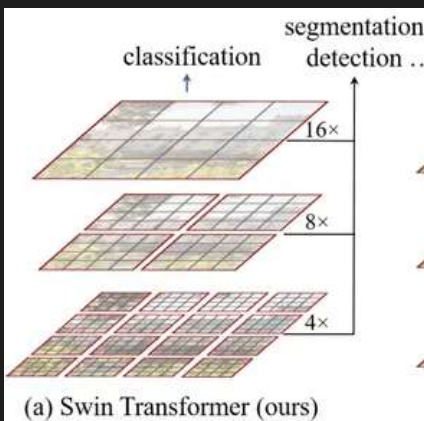
Masked MSA

Masked MSA

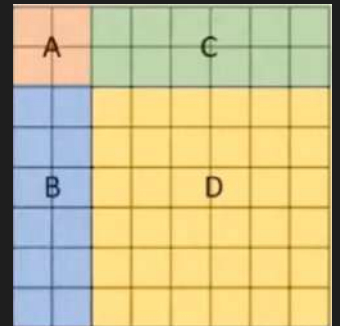
Masked MSA

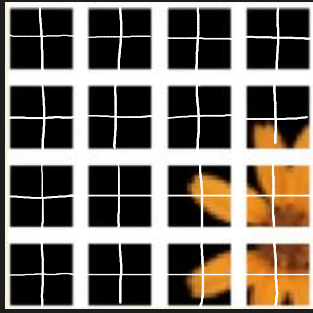
Reverse Cyclic Shift.

Patch Merging

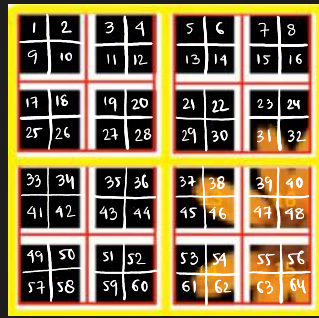


etc of these are in same window
here A can communicate with valid neighboring tokens if they are not blocked by attention mask

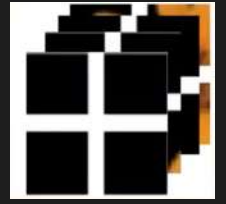
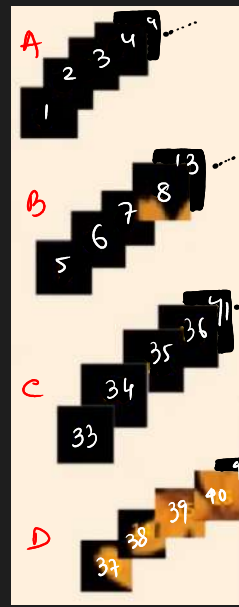




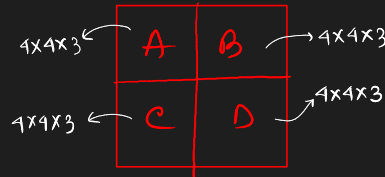
img: $8 \times 8 \times 3$
(RGB)



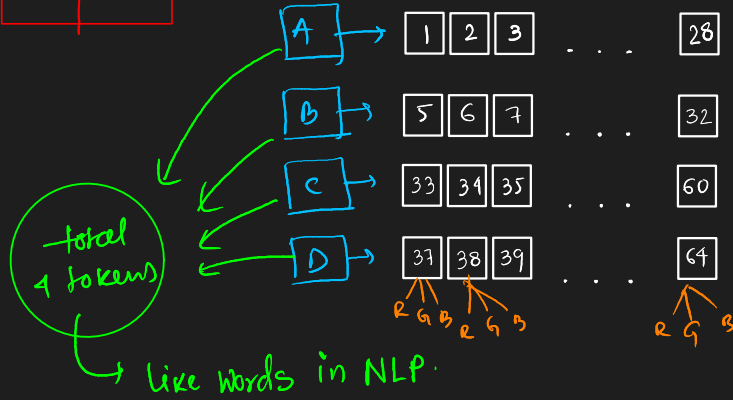
patch size = 4×4
total patches = 4.



4 tokens / patch.

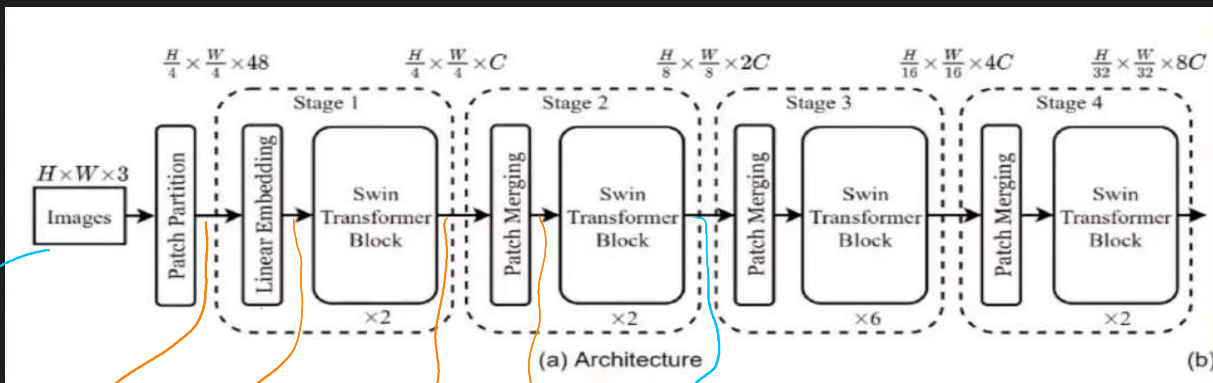
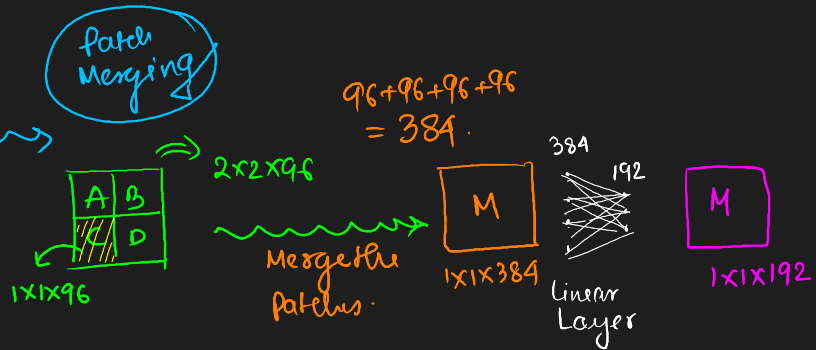
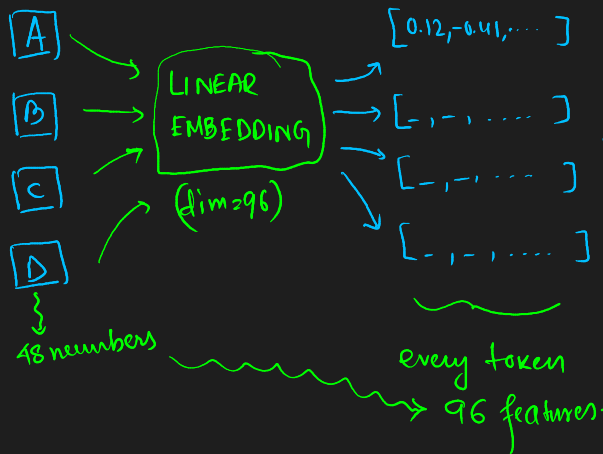


Flatten each patch



16 pixels each has 3 layers
(16 triplets in each token)
48 numbers after flattening.

Patches:



$8 \times 8 \times 3$

$$\frac{8}{4} \times \frac{8}{4} \times 48 = 2 \times 2 \times 48$$

$2 \times 2 \times 96$

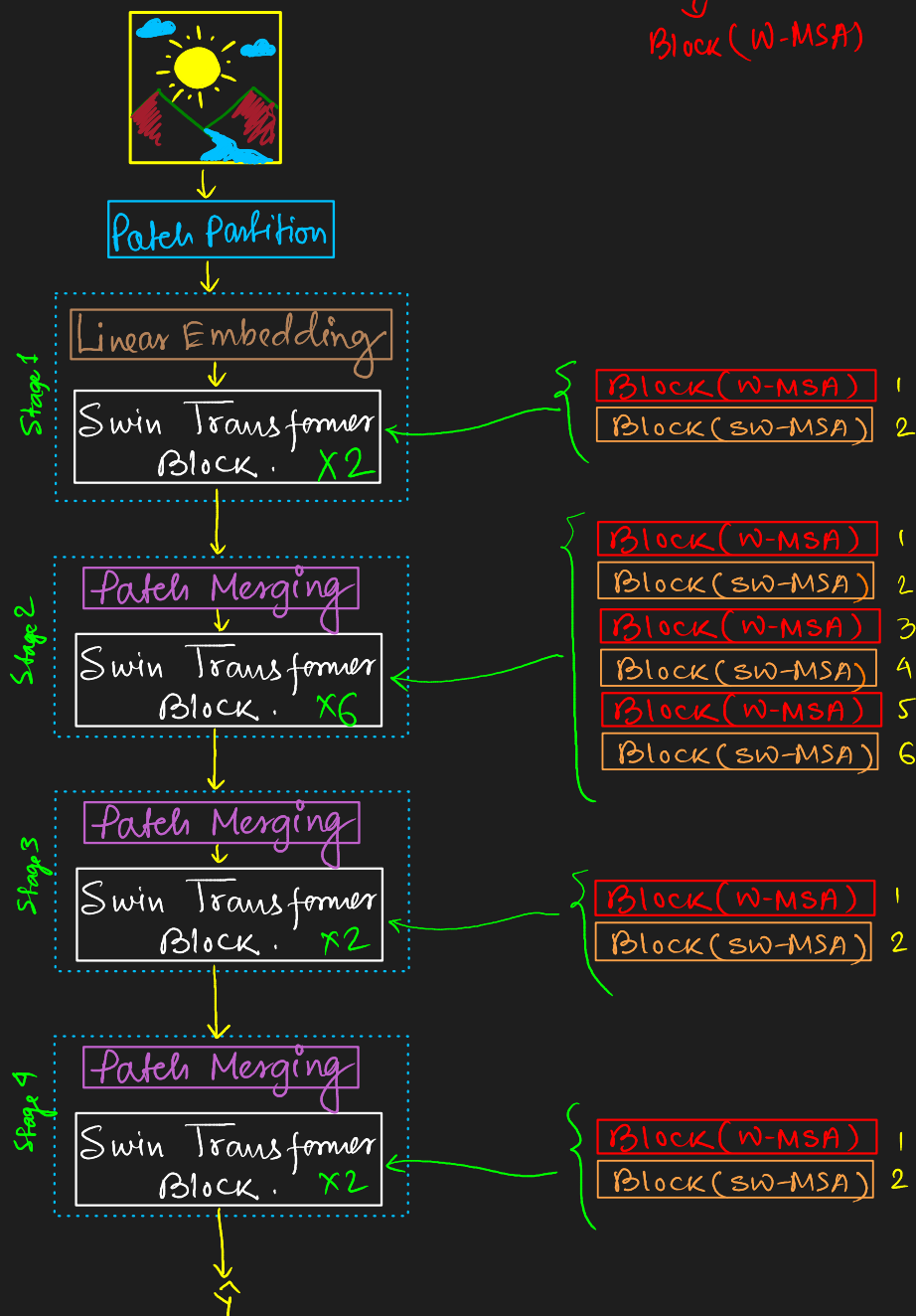
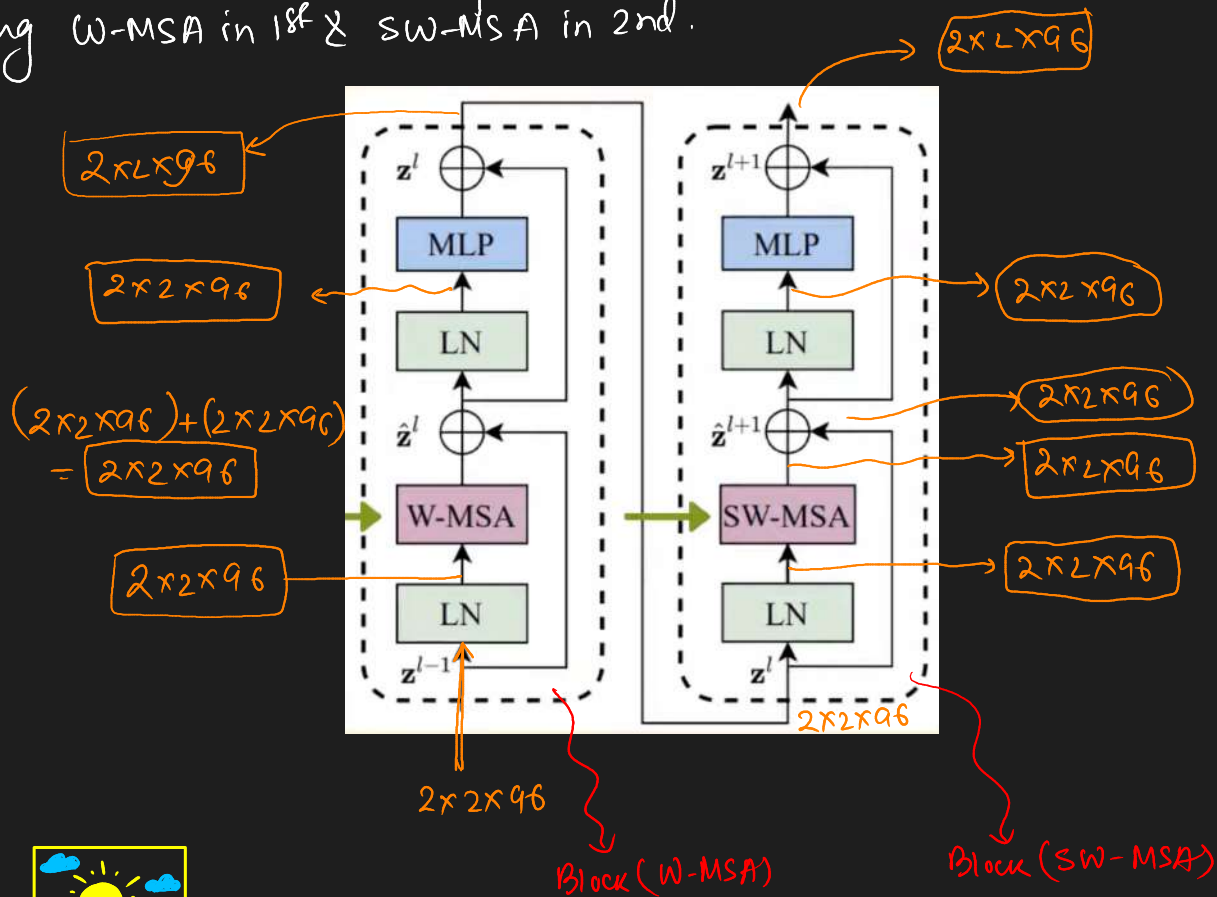
$2 \times 2 \times 96$

$1 \times 1 \times 192$

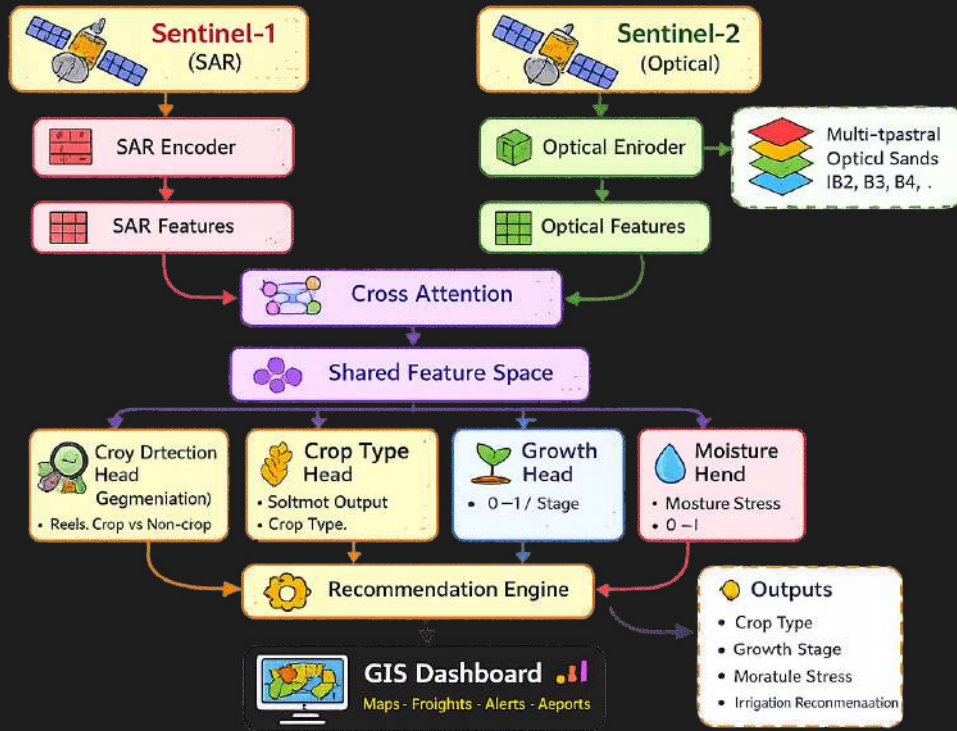
$1 \times 1 \times 192$

here further patch merging is not possible because the res ↓.

2 consecutive swin transformer block having W-MSA in 1st & SW-MSA in 2nd.



Our Idea



(VV, VH)
Sentinel-1
(SAR)

(B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12)
Sentinel-2
(OPTICAL)

(Patch=4x4)
(Embedding=96)

256x256x2

256x256x10

Patch Partition

Patch Partition

64x64x32

64x64x160

Stage 1

Stage 1

64x64x96

64x64x96

Stage 2

Stage 2

32x32x192

32x32x192

Stage 3

Stage 3

16x16x384

16x16x384

Stage 4

Stage 4

8x8x768

8x8x768

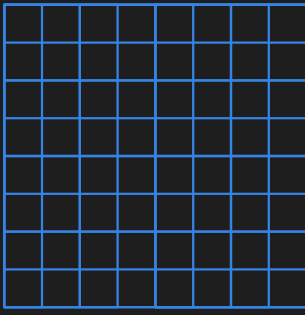
$$\frac{256}{4} = 64$$

$$4 \times 4 \times 10 = 160$$

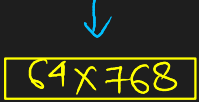
$$\frac{256}{4} = 64$$

$$4 \times 4 \times 2 = 32$$

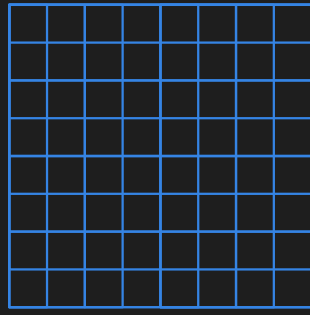
SAR
Feature
Map



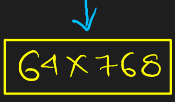
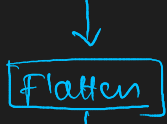
8x8x768



Optical
Feature
Map



8x8x768



From the problem statement:

- Optical data captures:
 - Vegetation vigor
 - Chlorophyll
 - Spectral signatures
 - Crop health
 - Growth stage
- SAR data captures:
 - Soil moisture
 - Crop structure
 - Surface roughness
 - Works through clouds
 - Day/night imaging

So the two modalities contain **complementary information**.

Which should be Query?

Your problem statement says:

"...using multi-source satellite data fusion for crop type classification, growth stage mapping, moisture stress detection..."

This means **neither modality is inherently more important**. Both are valuable.

However, if you must choose one direction for a **single Cross Attention**, you should justify it based on what you want to enhance.

Option 1 (Recommended for your problem)

Q = SAR

K = Optical

V = Optical

Reason:

SAR already provides reliable imaging under all weather conditions.

Optical contains richer semantic information:

- crop type
- vegetation indices
- phenology
- spectral characteristics

So SAR asks:

"I'm seeing this field from radar. Which optical features correspond to it?"

The output becomes

SAR + Optical information

This is especially useful because the final tasks are

- Crop Type
- Growth Stage
- Moisture Stress

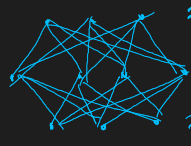
and Optical contributes strongly to Crop Type and Growth Stage.

The complementary option can also be taken.

SAR



Linear/Dense Layer



WQ

768

768

Q

Optical



K



V

64x768 Q

64x768 K

64x768 V

$$\text{Softmax}\left(\frac{QK^T}{\sqrt{768}}\right)V$$

64x768

Residual Conn.
LN
MLP
Residual Conn.
LN

64x768

shared feature space

Crop Presence Head

Binary Classification

0/1

if yes

BCE Loss

Crop Type Head

Multi class Classification

- Wheat
- Rice
- Maize
- Cotton
- etc.

Cross Entropy

Growth Stage Head

Multi Class Classification

- Germination
- Seeding
- Vegetative
- Flowering
- Fruiting

Cross Entropy

Moisture Head

Regression

0 → 100%

Recommendation